

AN INTRODUCTION TO STOCHASTIC PROCESSES: MARKOV CHAINS

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PAST PERFORMANCE IS NOT INDICATIVE ON FUTURE PERFORMANCE!

In the real world, many systems evolve in a way that is influenced by randomness, for example, asset prices, queueing systems, or if you want to come in from a physics standpoint, perhaps the movement of particles. These systems are commonly modelled using stochastic processes, which are collections of random variables indexed by time.

One of the most important classes of stochastic processes is a Markov chain. A Markov chain is characterised by the fact that its future behaviour depends only on its present state and not its past history. Despite the simple assumption, Markov chains can be incredibly effective in modelling for a wide range of real world events.

STOCHASTIC PROCESSES

A stochastic process can be discrete time or continuous time. A discrete time stochastic process is an infinite collection of random variables $X_0, X_1, \dots$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For example, think of the random variable, X_n , as being the value of an asset at the end of the n^{th} day after some specific date. Just before that date the value of the asset was given by X_0 . Thus, for any $\omega \in \Omega$, the sequence $X_0(\omega), X_1(\omega), \dots$ is a path of the evolution of the quantity described by the random variables X_n over time.

There are a couple of key definitions: a family, $\{\mathcal{F}_n\}_{n=0,1,2,\dots}$, of sub- σ -algebras of $\mathcal{F}$ is said to be a filtration iff it is increasing, i.e., $\forall n \leq m$, we have $\mathcal{F}_n \subseteq \mathcal{F}_m$. Very often, if $X = \{X_n\}_{n=0,1,2,\dots}$ is a stochastic process, we define a filtration by setting $\mathcal{F}_n = \sigma(\{X_m\}_{m \leq n})$. The σ -field, $\mathcal{F}_n$, represents the information that we have observed up to time n . This is called the canonical filtration of X . Now, let $\{\mathcal{F}_n\}_{n=0,1,2,\dots}$ be a filtration, then a stochastic process, Y , is said to be adapted to the filtration iff $\forall n$, the random variable Y_n is $\mathcal{F}_n$ -measurable.

MARKOV CHAIN

Consider a discrete-time stochastic process, $\{X_n\}_{n=0,1,2,\dots}$, where X_n takes values in the finite set $S = \{1, 2, \dots, N\}$ or $\{0, 1, \dots, N-1\}$. The possible values for X_n are called the states of the system. To describe the probabilities for such a process we need to give the values of $\mathbb{P}\{X_0 = i_0, X_1 = i_1, \dots, X_n = i_n\}$ for every n and every finite sequence of states $(i_0, i_1, \dots, i_n)$, this is the path. Equivalently, we could give the initial probability distribution and the transition probabilities, $q_n(i_n | i_0, i_1, \dots, i_{n-1}) = \mathbb{P}\{X_n = i_n | X_0 = i_0, X_1 = i_1, \dots, X_{n-1} = i_{n-1}\}$, $i = 1, 2, \dots, N$, thus, $\mathbb{P}\{X_0 = i_0, X_1 = i_1, \dots, X_n = i_n\} = \phi(i_0)q_1(i_1 | i_0)q_2(i_2 | i_0, i_1) \cdots q_n(i_n | i_0, i_1, \dots, i_{n-1})$. Informally, the Markov property states that to make predictions of the behaviour of a system in the future, we must consider only the present state of the system and not the past history. This is also known as the memoryless property. Mathematically, that is, $\mathbb{P}\{X_n = i_n | X_0 = i_0, X_1 = i_1, \dots, X_{n-1} = i_{n-1}\} = \mathbb{P}\{X_n = i_n | X_{n-1} = i_{n-1}\}$. We make the assumption that the transition probabilities do not depend on time, this is known as time homogeneity and mathematically represented by $\mathbb{P}\{X_n = i_n | X_{n-1} = i_{n-1}\} = \mathbb{P}\{X_1 = i_n | X_0 = i_{n-1}\}$, i.e., time homogeneity means that the probability of moving from state i to j is the same at every time step.

A time-homogenous Markov Chain is a process such that $\mathbb{P}\{X_n = i_n | X_0 = i_0, X_1 = i_1, \dots, X_{n-1} = i_{n-1}\} = p(i_{n-1}, i_n)$ for some function $p : S \times S \rightarrow [0, 1]$. To give the probabilities for a Markov chain, we must have an initial probability distribution, $\phi(i) = \mathbb{P}\{X_0 = i\}$, and transition probabilities, $p(i, j)$, thus, the probability of a path is $\mathbb{P}\{X_0 = i_0, X_1 = i_1, \dots, X_n = i_n\} = \phi(i_0)p(i_0, i_1)p(i_1, i_2) \cdots p(i_{n-1}, i_n)$, where $\mathbb{P}\{X_1 = i_1, \dots, X_n = i_n | X_0 = i_0\} = p(i_0, i_1)p(i_1, i_2) \cdots p(i_{n-1}, i_n)$. The transition matrix, P , for the Markov chain is the $N \times N$ matrix whose (i, j) entry, P_{ij} , is $p(i, j)$. The matrix, P , is a stochastic matrix, i.e., $0 \leq P_{ij} \leq 1, 1 \leq i, j \leq N$

and $\sum_{j=1}^N P_{ij} = 1, 1 \leq i \leq N$. A convenient way to visualise a Markov chain is through its transition graph. The vertices of the graph correspond to the states of the chain, and a directed edge is drawn from state i to state j whenever $p(i, j) > 0$. The edge is labelled with the transition probability, $p(i, j)$. The transition graph provides a graphical representation of the transition matrix, P . States which can be reached in a single step from state i correspond to the non-zero entries in the i^{th} row of P .

EXAMPLES: A TWO-STATE MARKOV CHAIN AND EVOLVED

Let us evaluate a simple model for the state of a phone where $X_n = 0$ means that the phone is free at time n and $X_n = 1$ means that the phone is busy at time n . Assume that during each time interval there is probability p that a call comes in, for ease assume that no more than one call comes in during any particular time interval. If the phone is busy during a time interval, there is probability q that it will be free during the next interval.

What Markov chain does the model give?

Solution: The model gives a Markov chain with state space $S = \{0, 1\}$ and transition matrix:

$$P = \{P_{ij}\} = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix} \end{matrix} = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}.$$

This matrix is the general form for a transition matrix of a two-state Markov chain. To specify the matrix, you only need the values of p and q . The matrix is written in two ways, first labels the states and the second doesn't.

Now modify the above example by assuming that the phone system can put one caller on hold. That is, at any time the number of callers in the system is in the set $S = \{0, 1, 2\}$. What Markov chain does the model give?

Solution: Set $p(0,0) = 1-p, p(0,1) = p, p(0,2) = 0$, since a caller comes in with probability p . Also, $p(2,0) = 0, p(2,1) = q, p(2,2) = 1-q$, since no new callers may arrive if there are two callers in the system. If there is only one caller in the system, then the state of the system can go from 1 to 0 if the current call is completed and no new callers enter the system, i.e., $p(1,0) = q(1-p)$. Similarly, the state goes from 1 to 2 if the current call is not completed but a new call arrives, i.e., $p(1,2) = p(1-q)$. Since the rows must add to 1, $p(1,1) = 1 - q(1-p) - p(1-q)$ and thus, the transition matrix is:

$$P = \{P_{ij}\} = \begin{bmatrix} 1-p & p & 0 \\ q(1-p) & 1 - q(1-p) - p(1-q) & p(1-q) \\ 0 & q & 1-q \end{bmatrix}.$$

CONCLUSION

Markov chains can be a simple yet powerful foundation for modelling systems that evolve randomly over time. By assuming that the future depends only on the present state, Markov chains allow complex probabilistic behaviour to be described through transition probabilities and matrices.

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